Part 1: Launch the DEVASC VM
If you have not already completed the Lab - Install the Virtual Machine Lab Environment, do so now. If you
have already completed that lab, launch the DEVASC VM.
Part 2: Review the Python Installation
These commands give you basic methods for learning more about the local Python environment.
a. In the DEVASC VM, you can check the version of Python already installed with the python3 -V
command.
devasc@labvm:~$ python3 -V
devasc@labvm:~$ which python3

```
devasc@labvm:~$ python3 -V
Python 3.8.10
devasc@labvm:~$ which python3
/usr/bin/python3
```

Part 3: PIP and Python Virtual Environments
Step 1: Create a Python 3 virtual environment.
a. Inside the DEVASC VM, change the labs/devnet-src/python directory.
devasc@labvm:~$ cd labs/devnet-src/python/
devasc@labvm:~/labs/devnet-src/python$ python3 -m venv devfun

```
devasc@labvm:~$ cd ~/labs/devnet-src/python/
devasc@labvm:~/labs/devnet-src/python$ python3 -m venv devfun
```

Step 2: Activate and test the Python 3 virtual environment.
a. Activate the virtual environment. The prompt changes to indicate the name of the environment you are
currently working in, which is devfun in this example. Now when you use the pip3 install command form
here, the system will only install packages for the active virtual environment.
devasc@labvm:~/labs/devnet-src/python$ source devfun/bin/activate

(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze

(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 install requests

```
devasc@labvm:~/labs/devnet-src/python$ source devfun/bin/activate
(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze
(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 install requests
Collecting requests
  Downloading requests-2.32.4-py3-none-any.whl (64 kB)
     |████████████████████████████████| 64 kB 502 kB/s
Collecting charset_normalizer<4,>=2
  Downloading charset_normalizer-3.5.2-cp37-abi3-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (253 kB)
     |████████████████████████████████| 253 kB 1.0 MB/s
Collecting urllib3<3,>=1.21.1
  Downloading urllib3-2.2.3-py3-none-any.whl (126 kB)
     |████████████████████████████████| 126 kB 2.0 MB/s
Collecting certifi>=2017.4.17
  Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)
     |████████████████████████████████| 136 kB 995 kB/s
Collecting idna<4,>=2.5
  Downloading idna-3.15-py3-none-any.whl (72 kB)
     |████████████████████████████████| 72 kB 391 kB/s
Installing collected packages: charset-normalizer, urllib3, certifi, idna, requests
Successfully installed certifi-2026.7.22 charset-normalizer-3.5.2 idna-3.15 requests-2.32.4 urllib3-2.2.3
```

Re-enter the pip3 freeze command to see the packages now installed in the devfun environment.
Note: Your list of packages and version numbers may be slightly different.
(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze

(devfun) devasc@labvm:~/labs/devnet-src/python$ deactivate

```
(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze
certifi==2026.7.22
charset-normalizer==3.5.2
idna==3.15
requests==2.32.4
urllib3==2.2.3
(devfun) devasc@labvm:~/labs/devnet-src/python$ deactivate
devasc@labvm:~/labs/devnet-src/python$
```

Step 3: Check the current packages installed in the system environment.
a. Enter the system wide python3 -m pip freeze command to see what packages are installed in the
system environment.
Note: Because Python 3 is invoked with the following command, you only use pip instead of pip3.
devasc@labvm:~/labs/devnet-src/python$ python3 -m pip freeze

devasc@labvm:~/labs/devnet-src/python$ python3 -m pip freeze | grep requests

```
devasc@labvm:~/labs/devnet-src/python$ python3 -m pip freeze
aiohttp==3.6.2
ansible==2.9.9
apache-libcloud==2.8.0
appdirs==1.4.3
apturl==0.5.2
argcomplete==1.8.1
arrow==0.15.6
astroid==2.3.3
async-timeout==3.0.1
attrs==19.3.0
backports.ssl==0.0.9
bcrypt==3.1.7
binaryornot==0.4.4
blinker==1.4
Brlapi==0.7.0
cajarename==19.7.15
certifi==2019.11.28
cffi==1.14.0
chardet==3.0.4
click==7.1.2
command-not-found==0.3
configobj==5.0.6
cookiecutter==1.7.2
cryptography==2.8
Menu  [Cisco DevNet: APIs, S...  devasc@labvm: ~/labs...
```

```
devasc@labvm:~/labs/devnet-src/python$ python3 -m pip freeze | grep requests
requests==2.22.0
requests-kerberos==0.12.0
requests-ntlm==1.1.0
requests-toolbelt==0.9.1
requests-unixsocket==0.2.0
```

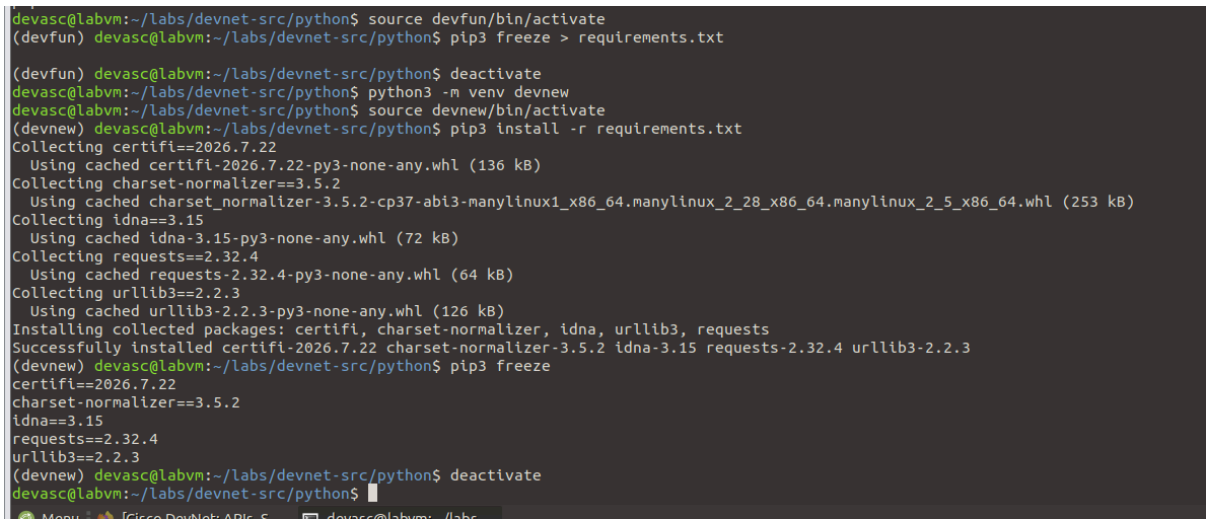
Part 4: Sharing Your Virtual Environment
Re-activate the devfun virtual environment.
devasc@labvm:~/labs/devnet-src/python$ source devfun/bin/activate

(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze >

(devfun) devasc@labvm:~/labs/devnet-src/python$ deactivate
devasc@labvm:~/labs/devnet-src/python$ ls

devasc@labvm:~/labs/devnet-src/python$ python3 -m venv devnew
devasc@labvm:~/labs/devnet-src/python$ source devnew/bin/activate
(devnew) devasc@labvm:~/labs/devnet-src/python$ pip3 install -r requirements.txt

(devnew) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze
(devnew) devasc@labvm:~/labs/devnet-src/python$ deactivate


devasc@labvm:~/labs/devnet-src/python$ source devfun/bin/activate
(devfun) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze > requirements.txt
(devfun) devasc@labvm:~/labs/devnet-src/python$ deactivate
devasc@labvm:~/labs/devnet-src/python$ python3 -m venv devnew
devasc@labvm:~/labs/devnet-src/python$ source devnew/bin/activate
(devnew) devasc@labvm:~/labs/devnet-src/python$ pip3 install -r requirements.txt
Collecting certifi==2026.7.22
Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)
Collecting charset-normalizer==3.5.2
Using cached charset_normalizer-3.5.2-cp37-abi3-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (253 kB)
Collecting idna==3.15
Using cached idna-3.15-py3-none-any.whl (72 kB)
Collecting requests==2.32.4
Using cached requests-2.32.4-py3-none-any.whl (64 kB)
Collecting urllib3==2.2.3
Using cached urllib3-2.2.3-py3-none-any.whl (126 kB)
Installing collected packages: certifi, charset-normalizer, idna, urllib3, requests
Successfully installed certifi-2026.7.22 charset-normalizer-3.5.2 idna-3.15 requests-2.32.4 urllib3-2.2.3
(devnew) devasc@labvm:~/labs/devnet-src/python$ pip3 freeze
certifi==2026.7.22
charset-normalizer==3.5.2
idna==3.15
requests==2.32.4
urllib3==2.2.3
(devnew) devasc@labvm:~/labs/devnet-src/python$ deactivate
devasc@labvm:~/labs/devnet-src/python$